

Start here x *pc problem.c x

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int mutex = 1;
5  int full = 0;
6  int empty = 3;
7  int x = 0;
8
9  void wait_sem(int *s)
10 {
11     (*s)--;
12 }
13
14
15 void signal_sem(int *s)
16 {
17     (*s)++;
18 }
19
20
21 void producer()
22 {
23     if ((mutex == 1) && (empty != 0))
24     {
25         wait_sem(&mutex);
26         full++;
27         empty--;
28         x++;
29
30         printf("Producer has produced: Item %d\n", x);
31
32         signal_sem(&mutex);
33     }
34     else
35     {
36         printf("Buffer is full!\n");
37     }
38 }
39
40
41 void consumer()
42 {
43     if ((mutex == 1) && (full != 0))
44     {
45         wait_sem(&mutex);
46         full--;
47         empty++;
48     }
```

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```
47     empty++;
48
49     printf("Consumer has consumed: Item %d\n", x);
50
51     x--;
52
53     signal_sem(&mutex);
54 }
55 else
56 {
57     printf("Buffer is empty!\n");
58 }
59 }
60
61 int main()
62 {
63     int choice;
64
65     while (1)
66     {
67         printf("\n1. Producer");
68         printf("\n2. Consumer");
69         printf("\n3. Exit");
70
71         printf("\nEnter your choice: ");
72         scanf("%d", &choice);
73
74         switch (choice)
75         {
76             case 1:
77                 producer();
78                 break;
79
80             case 2:
81                 consumer();
82                 break;
83
84             case 3:
85                 exit(0);
86
87             default:
88                 printf("Invalid choice!\n");
89         }
90     }
91
92     return 0;
93 }
94
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 1
Producer has produced: Item 1
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 1
Producer has produced: Item 2
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 1
Producer has produced: Item 3
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 1
Buffer is full!
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 2
Consumer has consumed: Item 3
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 2
Consumer has consumed: Item 2
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 2
Consumer has consumed: Item 1
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 2
Buffer is empty!
```

```
1. Producer
2. Consumer
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 1
Producer has produced: Item 3
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 1
Buffer is full!
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 2
Consumer has consumed: Item 3
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 2
Consumer has consumed: Item 2
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 2
Consumer has consumed: Item 1
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 2
Buffer is empty!
```

```
1. Producer
2. Consumer
3. Exit
Enter your choice: 3
```

```
Process returned 0 (0x0)    execution time : 35.455 s
Press any key to continue.
```

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